Functional Features Inventory

Template: No. of Functional Features Included in the Solution	Maximum Marks: 5 Marks
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Functional Features Implementation Table:

Comprehensive catalog of all functional features built, tested, and demonstrated in the FoodVerse AI platform.

S. No	Module / Epic	Feature Description	Input / Trigger	Output Generated	Complexity	Status
1	Food Scanner	Visual food plate recognition & macro nutrition estimation	Meal photo + optional voice note	Dish name, calories, protein, carbs, fats, health score	High	Completed
2	Freshness Detector	Computer vision decay & browning index inspection	Produce image (fruit/vegetable)	Freshness score (1-100), shelf life days, storage advice	High	Completed
3	Hygiene Auditor	Commercial kitchen video PPE & sanitation audit	Kitchen video clip / image	Compliance grade (A-F), violations, corrective actions	High	Completed
4	Cooking Assistant	Adaptive recipe generation & hands-free culinary Q&A;	Pantry ingredients / voice query	Step-by-step cooking guide with pro-tips and safety notes	Medium	Completed
5	Meal Planner	Metabolic BMR/TDEE meal planning with allergy safeguards	User biometrics, goal, allergies	Personalized daily meal plan, macro breakdown, grocery list	High	Completed
6	Synthetic Studio	HTML5 canvas synthetic test image generator	Preset buttons (apple, banana, plate)	Instant JPEG blob for rapid headless testing	Medium	Completed
7	Voice Transcriber	Microphone audio recording & Groq Whisper transcription	User speech through web mic	Clean transcribed text context for multimodal prompt	Medium	Completed

Feature Implementation Summary:

Metric	Count / Value	Evaluation Remarks
Total Features Proposed	7 Core Features	100% of planned architectural features scoped in requirements.
Total Features Implemented	7 Core Features	All features fully coded and integrated into FastAPI and React.
Total Features Demonstrated	7 Core Features	Live and accessible on https://foodverse-ai-six.vercel.app/.
Overall Implementation Rate	100.0%	Complete implementation with zero placeholder or mock endpoints.